

MAT 380: Exam 2

April 10, 2026

Name: _____

I wish you the best of luck! Some things to keep in mind with the exam:

- Please use complete sentences and proper formatting/notation when writing proofs. Reference any named theorem or axiom you use by name.
- You are encouraged to use scratch paper for scratch work, but I will not grade your scratch work.
- You can get partial credit for writing definitions/what you know about a problem if you don't know how to solve it.
- Pay attention to your points total and progress. Answer questions you know how to solve first, and move on if you are struggling. No question is worth more than 10 points.

If you have questions, ask Dr. Saeed to call me, I'll be available over phone.

The exam will take **1 hour** and is **60 total points**.

1. (15 points total, 3 points each) For the following sets, **determine if they are open, closed, both, or neither**. If they are not closed, **find their closure**. Also, answer if the set has any **isolated points**. No justification needed.

(a) $\emptyset$

Both. It has no points so it has no isolated points.

(b) $\mathbb{Q}$

Neither, $\overline{\mathbb{Q}} = \mathbb{R}$. No isolated points.

(c) $A = \{\frac{1}{n} : n \in \mathbb{N}\}$

Neither, $\overline{A} = A \cup \{0\}$. All points are isolated.

(d) $B = \{(-1)^n + \frac{2}{n} : n \in \mathbb{N}\}$

Neither, $\overline{B} = B \cup \{-1\}$. All points are isolated except 1.

(e) Cantor “Middle Thirds” set C

Closed, not open. No isolated points.

2. (20 points total, 5 points each) Decide whether each statement is **true or false**. Provide a justification for those that are true and a counterexample for those that are false (with a brief explanation of why your counterexample works).

(a) The arbitrary union of compact sets is compact.

Not true. Consider $\bigcup_{n=1}^{\infty} [n, n+1] = [1, \infty)$ which is not bounded so is not compact.

(b) For a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$, if $g(x_0) > 0$ for a single point $x_0 \in \mathbb{R}$, then $g(x)$ is in fact strictly positive for uncountably many points.

True. Since $g(x_0) > 0$, let $\epsilon = g(x_0)$. Then there is a $\delta > 0$ such that all $x \in V_\delta(x_0)$ have $f(x) \in V_\epsilon(f(x_0))$.

(c) If $f(K)$ is compact for any compact set $K \subseteq \mathbb{R}$, then $f : \mathbb{R} \rightarrow \mathbb{R}$ must be continuous on its domain.

Not true. Consider Dirichlet's function. The image of any set, let alone any compact set, is either $\{0\}$, $\{1\}$, or $\{0, 1\}$, all three of which are compact. However Dirichlet's function is not continuous anywhere on its domain.

(d) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then if A is bounded, then $f^{-1}(A)$ must be bounded.

Not true. Consider the constant function $f(x) = 0$. It is clearly continuous, but the preimage of the bounded set $\{0\}$ is $\mathbb{R}$, which is not bounded.

3. (5 points) Prove that if a set G is open, then its complement G^c is closed.

Proof. We need to show that G^c contains all of its limit points.

Let x be an arbitrary limit point of G^c .

Then for all $\epsilon > 0$, $V_\epsilon(x)$ intersects G^c somewhere.

So there is no $\epsilon > 0$ such that $V_\epsilon(x) \subseteq G$.

Since G is open, $x \notin G$. So $x \in G^c$.

□

4. (10 points) Prove the Preservation of Compact Sets using the sequential definitions of compactness and continuity.

Thm: Let $f : A \rightarrow \mathbb{R}$ be continuous on A . If $K \subseteq A$ is compact, then $f(K)$ is compact.

Proof. Let $K \subseteq A$ be an arbitrary compact set. We need to show $f(K)$ is compact.

Let (y_n) be an arbitrary sequence in $f(K)$.

The preimage $f^{-1}(\{y_n\}) \in K$. Choose $x_n \in f^{-1}(\{y_n\})$. This is a sequence $(x_n) \in K$.

Since K is compact, it has a convergent subsequence $(x_{n_k}) \rightarrow x$, where $x \in K$.

Since f is continuous, $(f(x_{n_k})) \rightarrow f(x)$ where $f(x) \in f(K)$.

$(f(x_{n_k}))$ is a subsequence of (y_n) that converges to a limit in $f(K)$.

So by the sequential definition of compactness, $f(K)$ is compact. □

5. (10 points) Assume a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on its domain. Prove that every open set $O \subseteq \mathbb{R}$ has an open preimage, that is, $f^{-1}(O)$ is an open set as well.

Proof. Assume that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on its domain and $O \subseteq \mathbb{R}$ is an open set.

Show $f^{-1}(O)$ is an open set.

Choose some arbitrary element $c \in f^{-1}(O)$.

Then $f(c) \in O$.

Since O is open, there exists $V_\varepsilon(f(c)) \subseteq O$.

Since f is continuous, there exists some $V_\delta(c)$ such that if $x \in V_\delta(c)$, we have $f(x) \in V_\varepsilon(f(c))$ (and therefore $f(x) \in O$).

So then $V_\delta(c) \subseteq f^{-1}(O)$, and therefore $f^{-1}(O)$ is open by definition. □